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absolute difference within 5 mmHg and 82 out of 99 for diastolic blood pressure (DBP). As for Omron M6 Comfort®, a total of 83 out of 99 comparisons for SBP showed an absolute difference within 5 mmHg and 77 out of 99 for DBP. The mean differences between the device and mercury readings were 0.7±5.9 mmHg for SBP and 0.3±4.1 mmHg for DBP for QardioArm® and −1.4±4.7 mmHg for SBP and −2.1±4.3 mmHg for DBP for Omron M6 Comfort®. With regard to part 2 of ESH-IP 2010, 27 out of 33 subjects had a minimum of two out of three measurements within 5 mmHg difference for SBP and 31 out of 33 subjects for DBP for the QardioArm®, and 29 out of 33 patients had a minimum of two out of three measurements within 5 mmHg difference for SBP and 26 out of 33 patients for DBP for Omron M6 Comfort®.
Conclusion: QardioArm® and Omron M6 Comfort® readings differing from the mercury standard by <5, 10, and 15 mmHg fulfill the ESH-IP Revision 2010 requirements. Consequently, these two devices are suitable for use in the general population and non-insulin-dependent type II diabetic patients, respectively.

Keywords: QardioArm®, Omron M6 Comfort®, validation, blood pressure measurement, type II diabetes, International Protocol

Introduction

Over the last years, many automated blood pressure (BP) measuring devices have been developed to gradually supplant the standard mercury sphygmomanometer due to mercury limitations.[1,2] A validation process for these devices is therefore needed to verify their accuracy following different protocols, such as the European Society of Hypertension International protocol (ESH-IP),[3] the British Hypertension Society (BHS) protocol,[4] and the Association for the Advancement of Medical Instrumentation (AAMI) protocol.[5] This validation process is required in the general population and in specific populations such as type II diabetic patients, for example, where accurate BP measurement is crucial, so the effective management of hypertension can delay disease complications.

Type II diabetes is a major health problem that has higher prevalence in the older population.[6,7] It is well known that atherosclerotic disease is the major complication of type II diabetes.[8,9] Atherosclerosis leads to the degeneration of arterial elasticity and subsequent arterial wall stiffening which is a strong risk factor for early mortality in patients with type II diabetes.[2,10] Arterial wall stiffening is exacerbated with diabetes[11] and hypertension.[10,12–15] It is estimated that high BP occurs in over two-thirds of patients with type II diabetes.[16] Recent evidence has shown that BP control decreases the risk of developing diabetes-related microvascular, and microvascular complications.[16] Knowing that arterial stiffness is more critical in type II diabetic patients with hypertension than those without hypertension,[11] and that increase in arterial stiffness may affect the accuracy of automatic BP measurements,[11,17] accurate BP measurements in such populations is therefore crucial for the diagnosis and management. However, only a limited number of studies have tested the accuracy of automated BP monitors in the type II diabetic population,[18–21] and the majority of these validations did not adhere to existing validation standards according to specific protocols cited above.[3–5] Therefore, the issue of BP measurements accuracy remains controversial.[22] Because of the high prevalence of type II diabetes, and the importance to reduce morbidity and mortality associated to hypertension, it is important to assess the accuracy of automatic BP measurements in this specific population. Most importantly, measuring BP accurately will facilitate the establishment of a threshold level of BP above which antihypertensive treatment can be recommended especially in type II diabetic patients, highly prone to developing hypertension and its complications if misdiagnosed.

In 2014, the Omron M6 Comfort® BP device was only validated in the general population following the ESH-IP;[24] and QardioArm® device has never been validated previously in any population. The aim of this study is to check the validity of the QardioArm® BP measurement device in the general population and Omron M6 Comfort® oscillometric BP measurement device in non-insulin-dependent type II diabetic patients, following the ESH-IP 2010.

Materials and methods

Ethical information

The study was approved by the local ethic committees, namely National Institute of Health, Ministry of Health, Republic of Armenia and Institutional Review Board and Ethical Committee Geitaoui Hospital, Beirut, Lebanon.

All eligible subjects included in this study signed an informed consent form prior to any BP measurement.

Tested devices

The QardioArm® device The QardioArm® device, provided by Qardio Inc (San Francisco, CA, USA), is a home BP measurement automatic device at the arm level. It needs a device with Bluetooth 4.0, iOS 7.0 (or later), Android 4.4 "KitKat" (or later), and is compatible with iPhone, iPod, iPad, and Apple Watch as well as with Android phones and tablets. In order to operate the device, a specific free Qardio application can be downloaded from the Apple App Store or Google Play (or www.getqardio.com). The unit records BP using the oscillometric method, with an automatic inflation and a controlled pressure release valve, with a pressure range of 40–250 mmHg and pulse rate range of 40–200 beats/min. The unit weighs ~310 g including batteries. Four 1.5 V AAA batteries are needed. The included single cuff is easy-fit Lycra, applicable to arm circumferences ranging from 22 to 37 cm. The dimensions are 140×68×38 mm, when closed.

The Omron M6 IT Comfort® device

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This device (HEM-7322U-E) was provided and randomly selected by the manufacturer (Omron Healthcare, Kyoto, Japan). F
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Abbreviations: BP, blood pressure; DBP, diastolic blood pressure; ESH-IP, European Society of Hypertension International Protocol; SBP, systolic blood pressure; SD, standard deviation.

General population A total of 33 participants fulfilled the ESH-IP recruitment criteria and were included in the validation of the QardioArm® device. All participants' arm circumference was between 22 and 37 cm and none of them had atrial fibrillation or any other arrhythmia. Subjects were preselected from the general population in Yerevan, Armenia.

Non-insulin-dependent type II diabetic population A total of 33 participants fulfilled the ESH-IP recruitment criteria and were included in the validation of the Omron M6 IT Comfort® device. All participants' circumference was between 22 and 42 cm and none of them had atrial fibrillation or any other arrhythmia or used insulin as treatment for type II diabetes. Patients were preselected from outpatient clinics and from inpatients at the Lebanese University affiliated hospitals (Mount-Lebanon, Lebanese Hospital Geitaoui, Rafik Hariri University Hospital, Governmental Hospital of Baabda, Rasoul el-Aazam Hospital, and Bahman Hospital).

For BP measurement, three persons were included in the validation team: two observers trained in accurate BP measurement and a supervisor, all of them having completed a training based on a CD-ROM specifically developed by the French Society of Hypertension to certify observers involved in clinical studies and to familiarize them with the use of tested devices.

Each device validation study was assessed in different populations (general population/type II diabetic populations), separately from the other device validations and at another time. Therefore, each patient participated in only one study device validation. Evaluation of the devices was done according to the ESH-IP Revision 2010.

In our study, the two observers used two parallel connected mercury sphygmomanometers and a teaching stethoscope as a reference standard. They also measured the circumference of the arm to make certain that the cuff size being used is adequate for the subject. The agreement between the two observers, blinded from each other's result, was checked all over the evaluation period by the supervisor to ensure that the difference between the two observers is ≤4 mmHg for systolic and diastolic BP values. Otherwise, the measurement was repeated. Measurements through the mercury sphygmomanometer were performed according to the "same arm, consecutive measurements" supported at the heart level. Measurements by both devices were on the same arm supported at the heart level, as recommended by the manufacturers asking the subject to relax for 5–10 minutes, while making sure that the subject is seated with uncrossed legs and supported back. Two cuffs were used for the standard mercury sphygmomanometer measurements: standard cuff for 22–32 cm and large cuff for 32–42 cm arm circumference. In total, in each patient, nine consecutive BP measurements were performed using the mercury sphygmomanometers (five times) and the tested device (four times), and were recorded as follows:

- BPA: entry BP, observers 1 and 2 each with the mercury standard
- BPB: device detection BP, supervisor
- BP1: observers 1 and 2 with mercury standard
- BP2: supervisor with the test instrument
- BP3: observers 1 and 2 with mercury standard
- BP4: supervisor with the test instrument
- BP5: observers 1 and 2 with mercury standard
- BP6: supervisor with the test instrument
- BP7: observers 1 and 2 with mercury standard

Certainly, the interval between one BP measurement and the next one was 30–60 seconds.

Data analysis

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The accuracy of a device according to the ESH-IP is based on a comparison between the device and reference (mercury) measur
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Bland–Altman plots of the differences between BP measurements obtained with the QardioArm® device and the sphygmomanometer are shown for SBP (Figure 1A) and for DBP (Figure 1B). These results are in concordance with the requested criteria of the ESH-IP and the device succeeded in fulfilling the validation criteria of the ESH-IP Revision 2010.



Figure 1 Plots of systolic (**A**) and diastolic (**B**) BP differences between the QardioArm® BP device and the mean of two observers' readings in 33 subjects (n=99 measurements).

Notes: Delta SBP dev–ref: systolic difference between the QardioArm® and mercury sphygmomanometer; mean SBP dev–ref: mean systolic average values of the QardioArm® and the mercury sphygmomanometer. Delta DBP dev–ref: diastolic difference between the QardioArm® and mercury sphygmomanometer; mean DBP dev–ref: mean diastolic average values of the QardioArm® and the mercury sphygmomanometer.

Abbreviations: BP, blood pressure; DBP, diastolic blood pressure; dev, device; ref, reference; SBP, systolic blood pressure.

The Omron M6 IT Comfort® device

Patients' details In all, 58 non-insulin-dependent type II diabetic patients were screened to recruit 33 participants who met the ESH-IP criteria and 25 were excluded for reasons mentioned in Table 3A. The 33 remaining participants were recruited following the three different BP ranges, such as 11 subjects in each of the ranges in Table 3B. There were 14 males and 19 females; the mean age of the participants was 61±11 years. Their arm circumference ranged from 24 to 36 cm, with a total of 14/33 participants having an arm circumference ≥32 cm, where one universal rigid cuff (22–42 cm) with the Omron M6 IT Comfort® device was used, and the standard (22–32 cm) (n=19 patients) and large (32–42 cm) (n=14 patients) cuffs were used with the mercury sphygmomanometer, according to patients' arm size. The mean recruitment SBP was 143.8±24 mmHg (99–178 mmHg) and the mean recruitment DBP was 86.4±18 mmHg (48–121 mmHg). More details about participants' characteristics are described in Table 4A.

BP requirements Observer measurements in each recruitment range are described in Table 4B. The number of observer test measurements in each pressure range is, as required by the ESH-IP, between 22 and 44; more specifically, it is between 31 and 36 measurements. The difference between the range with the highest count and that with the lowest count is, as required by the ESH-IP, ≤19; more specifically, it is 5 for SBP and 4 for DBP. The overall SBP range is, as required by the ESH-IP, from ≤100 to ≥170 mmHg; more specifically, it varies from 98 to 175 mmHg. The overall DBP range is, as required by the ESH-IP, from ≤50 to ≥120 mmHg; more specifically, it ranges from 49 to 124 mmHg.

BP measurements The difference between the two observers was −0.2±2.2 and +0.3±2.1 mmHg for SBP and DBP, respectively (−4 to +4 mmHg). The mean differences between the observers and the tested device were −1.4±4.7 mmHg for SBP and −2.1±4.3 mmHg for DBP. The numbers of measurements differing from the mercury standard by 5, 10, and 15 mmHg or less are shown in Table 6. Indeed, a total of 83 out of 99 comparisons for SBP showed an absolute difference within 5 mmHg and 77 out of 99 for DBP (vs at least 73 for SBP and 65 for DBP following ESH-IP requirements). In addition, a total of 94 out of 99 comparisons for SBP showed an absolute difference within 10 mmHg and 93 out of 99 for DBP (vs at least 87 for SBP and 81 for DBP following ESH-IP requirements). Furthermore, a total of 97 out of 99 comparisons for SBP showed an absolute difference within 15 mmHg and 99 out of 99 for DBP (vs at least 96 for SBP and 93 for DBP following ESH-IP requirements). Since "two" out of these three absolute differences of 5 or 10 or 15 mmHg for SBP are within the required range mentioned above and "all" (three out of three) of them for DBP, part 1 device validation was successfully completed.



Table 6 Validation results for the Omron M6 IT Comfort® BP device according to the ESH-IP 2010

Notes: Pass Req: as required by the IP; achieved: as recorded by the device. Refer to the text for more explanations about parts 1–3.

Abbreviations: BP, blood pressure; DBP, diastolic blood pressure; IP, International Protocol; ESH-IP, European Society of Hypertension International Protocol; SBP, systolic blood pressure; SD, standard deviation.

As for part 2 of ESH-IP 2010, 29 out of 33 subjects had a minimum of two out of three measurements within 5 mmHg difference for SBP and 26 out of 33 subjects for DBP (vs at least 24 out of 33 subjects for SBP and DBP following ESH-IP requirements). In addition, none of the 33 subjects had a minimum of zero out of three measurements within 5 mmHg

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Finally, there is no explicit criteria mentioned for a validation process in specific populations in the ESH-IP, and this is highly recommended to be taken into consideration in its next revision. Actually, the presence of several protocols for validation such as AAMI,[5] BHS,[6] ESH-IP[2] is problematic for several reasons: 1) manufacturers cannot perform all three protocols; 2) experts are recognizing mainly their national protocols; for example, AAMI in the USA, ESH-IP in Europe; 3) impossibility to compare various studies; therefore, there is an ultimate need for a unified international protocol.[22]

Conclusion

This study showed that the QardioArm® device and Omron M6 Comfort® device successfully passed the validation requirements in the general population and in type II diabetic patients, respectively, according to the ESH-IP Revision 2010.

Acknowledgments

Qardio Inc and Omron Healthcare have financially supported this study. They initially provided three devices each; one of them was chosen randomly and selected for the study.

Disclosure

The authors report no conflicts of interest in this work.

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